REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 10 Advanced | Module 16 Lesson 4 | Convex and Concave Lenses

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: 1/f = 1/do + 1/di; m = hi/ho = −di/do. Convex f positive; concave f negative. Real images have positive di.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 10 book, printed pages 26–35.

P01. A convex lens in air is typically thicker:

A. At its centre than at its edges B. At its edges than at its centre
C. Only on one side of its frame D. By exactly one wavelength everywhere

Answer

A. At its centre than at its edges.

P02. A concave lens in air usually makes parallel rays:

A. Converge to a real focus B. Diverge
C. Stop completely D. Reflect backward along all paths

Answer

B. Diverge.

P03. A real object viewed through a single concave lens has an image that is:

A. Real inverted and enlarged B. Real upright and equal-sized
C. Virtual upright and diminished D. Virtual inverted and enlarged

Answer

C. Virtual upright and diminished.

P04. Nearsightedness is commonly corrected with a:

A. Convex lens B. Plane mirror
C. Prism alone D. Concave lens

Answer

D. Concave lens.

P05. Chromatic aberration results mainly from:

A. Different wavelengths having different focal lengths B. Light having no frequency
C. A lens having zero thickness everywhere D. Equal refraction for all colours

Answer

A. Different wavelengths having different focal lengths.

P06. A thin convex lens has f = +10 cm and do = +30 cm. Calculate its signed image distance.

A. 15 cm B. 30 cm
C. 7.5 cm D. -15 cm

Answer

A. 15 cm. 1/di = 1/f − 1/do = 1/10 − 1/30; di = +15 cm, a real image.

P07. A thin concave lens has f = -10 cm and do = +30 cm. Calculate its signed image distance.

A. -15 cm B. -7.5 cm
C. -3.75 cm D. 7.5 cm

Answer

B. -7.5 cm. 1/di = 1/(-10) − 1/30; di = -7.5 cm, a virtual image on the object side.

P08. A lens has object distance +24 cm and real image distance +12 cm. Calculate its signed magnification.

A. -1 B. -0.25
C. -0.5 D. 0.5

Answer

C. -0.5. m = −di/do = −12/24 = -0.5. The image is inverted and half the object height.

P09. A lens produces magnification −2.0 for an object 4 cm tall. Find its signed image height.

A. -16 cm B. -4 cm
C. 8 cm D. -8 cm

Answer

D. -8 cm. hi = mho = −2.0 × 4 = -8 cm. Negative indicates inversion.

P10. A lens forms a real image at 30 cm for an object 60 cm away. Calculate its focal length.

A. 20 cm B. 40 cm
C. 10 cm D. -20 cm

Answer

A. 20 cm. 1/f = 1/do + 1/di = 1/60 + 1/30; f = 20 cm.

P11. Define a lens.

Answer

A transparent optical element with refracting surfaces that redirects light to form images.

P12. Distinguish a real image from a virtual lens image.

Answer

A real image is formed by actual convergence of rays and can be projected onto a screen; a virtual image is formed by backward extensions of diverging rays.

P13. How is farsightedness commonly corrected?

Answer

A suitable convex lens increases convergence so near-object rays can be focused on the retina.

P14. Describe spherical aberration.

Answer

Rays far from the axis focus at a different position from paraxial rays, blurring the image.

P15. Why do compound achromatic lenses reduce chromatic aberration?

Answer

They combine materials and curvatures whose dispersions partly compensate, bringing selected colours to nearly the same focus.

P16. A thin convex lens has f = +15 cm and do = +45 cm. Calculate its signed image distance.

Answer

1/di = 1/f − 1/do = 1/15 − 1/45; di = +22.5 cm, a real image.

P17. A thin convex lens has f = +20 cm and do = +60 cm. Calculate its signed image distance.

Answer

1/di = 1/f − 1/do = 1/20 − 1/60; di = +30 cm, a real image.

P18. A thin concave lens has f = -15 cm and do = +45 cm. Calculate its signed image distance.

Answer

1/di = 1/(-15) − 1/45; di = -11.25 cm, a virtual image on the object side.

P19. A thin concave lens has f = -20 cm and do = +60 cm. Calculate its signed image distance.

Answer

1/di = 1/(-20) − 1/60; di = -15 cm, a virtual image on the object side.

P20. A lens has object distance +36 cm and real image distance +18 cm. Calculate its signed magnification.

Answer

m = −di/do = −18/36 = -0.5. The image is inverted and half the object height.

P21. A lens has object distance +48 cm and real image distance +24 cm. Calculate its signed magnification.

Answer

m = −di/do = −24/48 = -0.5. The image is inverted and half the object height.

P22. A lens produces magnification −2.0 for an object 6 cm tall. Find its signed image height.

Answer

hi = mho = −2.0 × 6 = -12 cm. Negative indicates inversion.

P23. A lens produces magnification −2.0 for an object 8 cm tall. Find its signed image height.

Answer

hi = mho = −2.0 × 8 = -16 cm. Negative indicates inversion.

P24. A lens forms a real image at 45 cm for an object 90 cm away. Calculate its focal length.

Answer

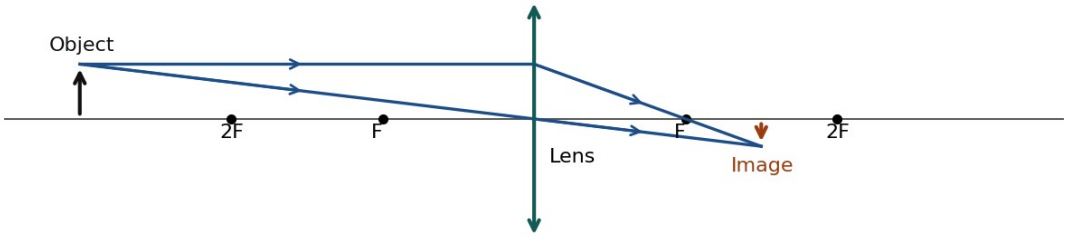
1/f = 1/do + 1/di = 1/90 + 1/45; f = 30 cm.

P25. A lens forms a real image at 60 cm for an object 120 cm away. Calculate its focal length.

Answer

1/f = 1/do + 1/di = 1/120 + 1/60; f = 40 cm.

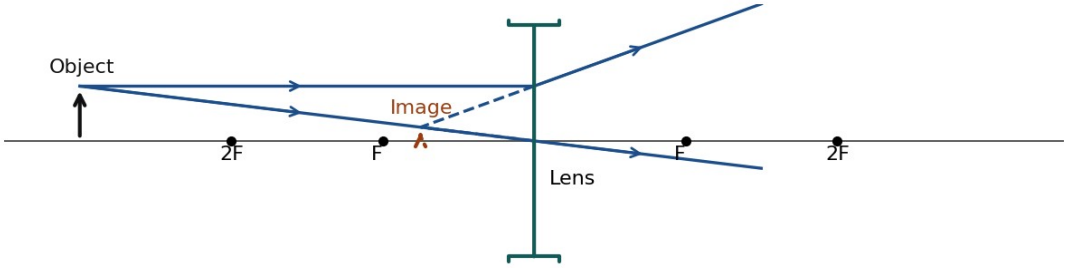
P26. Draw a ray diagram for an object beyond twice the focal length of a convex lens.



Answer

A ray parallel to the axis refracts through the far focus; a central ray continues approximately straight. The image is real, inverted and diminished between F and 2F on the far side.

P27. Draw the image of a real object through a concave lens and describe its properties.



Answer

Draw a parallel ray refracting as if from the near focus and a central straight ray. Backward extensions meet between lens and near focus: virtual, upright and diminished.

P28. Why can a convex lens serve as a magnifying glass when the object lies inside F?

Answer

The refracted rays diverge; their backward extensions form an upright enlarged virtual image on the object side.

P29. A student says a negative image distance always means the lens is concave. Evaluate the claim.

Answer

False. A convex lens also produces a negative image distance for an object inside its focal length. The sign describes a virtual image under the convention.

P30. Explain why a camera normally uses a real image rather than a virtual image at its sensor.

Answer

The sensor must receive an actual focused distribution of light. A real image forms where rays converge onto the sensor plane.